

David R. Maestas, Jr., PhD

Postdoctoral Associate, Department of Bioengineering, University of Pittsburgh
Immunoengineering · Regenerative Medicine · Vascular Tissue Engineering
Email: drm154@pitt.edu
Updated July 2026

EXECUTIVE SUMMARY

- Immunomechanical engineering researcher focused on wound repair mechanisms and tissue regeneration
- Specialized expertise in wound repair immunology, immuno-stromal polarization, & vascular remodeling
- Builds discovery platforms that leverage live 3D imaging modalities for longitudinal tracking
- Uses mechanistic insight to design regenerative therapies that tune inflammation, remodeling, & mechanics
- Emerging leader with strengths in mentorship, interdisciplinary collaboration, & experimental creativity

EDUCATION

Johns Hopkins University — *Baltimore, Maryland* *2016-2022*

Doctor of Philosophy: Biomedical Engineering

Focus Areas: Regenerative Medicine, Fibrosis, Immunoengineering

- National Science Foundation Graduate Research Fellowship Award
- Drescher Graduate Student Medical Research Award

The University of Arizona — *Tucson, Arizona* *2013-2016*

Bachelor of Science: Biomedical Engineering

Focus Areas: Regenerative Medicine, Cell & Tissue Engineering

- Danny Brower Scholar Award
- Galileo Circle Scholar Award

Pima Community College — *Tucson, Arizona* *2010-2013*

Associate of Science & General Studies: Biomedical Engineering w/Honors

Focus Areas: Biology and Engineering

- Governor's Board Special Recognition Scholar Award
- Analytical Math Association (AMATYC) Scholar Award

RESEARCH EXPERIENCE

Soft-Tissue Biomechanics Laboratory — *University of Pittsburgh* *Pittsburgh, PA*

Post-doctoral Associate: Dr. Jonathan Vande Geest

2022-Present

- Development of pre-clinical tissue engineered vascular grafts
- Investigation of biomechanical cues that impact vascular regeneration
- Immunoengineering of vascular grafts for tissue regeneration
- Development of a 3D Artery-TEVG *Ex Vivo* Culture System

Elisseeff Lab — *Johns Hopkins University* *Baltimore, MD*

Pre-Doctoral Researcher: Dr. Jennifer H. Elisseeff

2016-2022

- Development of pro-regenerative immunotherapies from helminth antigens
- Utilized multiple preclinical murine surgery models to assess wound repair
- Engineered screening assays for pro-regenerative materials and agonists

Tax Lab for Model Plant Genetics — *The University of Arizona* *Tucson, AZ*

Undergraduate Researcher: Dr. Frans Tax

2013-2016

- Genotyping to assess the roles of receptor-like kinases in embryonic patterning.
- Confocal image screening for validation of knock-in reporter fluorophores.

Soft-Tissue Biomechanics Laboratory — *The University of Arizona* *Tucson, AZ*

Undergraduate Researcher: Dr. Jonathan Vande Geest

2013-2016

- Developed vascular grafts seeded with umbilical cord blood derived stem cells
- Assessment of microstructure and thrombogenic activity of scaffolds

1. **Resolving differential vascular graft remodeling using longitudinal multiphoton tracking in a 3D culture platform.** Maestas, Jr., D.R. T.R. Murphy, K.M. Martinet, T. Moyston, L.X. Min, A. Behrangzade, B.J. Pemberton, J. Joy, S. Ye, G.S. Hussey, Mohamad Azhar, WR Wagner, and JP Vande Geest. *Materials Today Bio* (under review). DOI: [10.64898/2026.02.28.708759](https://doi.org/10.64898/2026.02.28.708759)
2. **Immune checkpoint inhibitors amplify type 2 immune mediated repair by pro-regenerative scaffolds.** Garcia, J., Ruta, A., Yu, F.H., Mejías, J.C., Peña, A.N., Rutkowski, N., Gray-Gaillard, E.F., Dubois, C., Cherry, C., Browne, M., Stivers, K., Maestas, D.R., Krishnan, K., Bell, A.T.F., Fertig, E.J., Cooney, C.M., Cooney, D., Byrne, P., Hillel, A.T., Smith, K.N., Ji, H., Anders, R., Pardoll, D.M. and Elisseeff, J.H., *bioRxiv*, 2026. DOI: [10.64898/2026.01.31.703034](https://doi.org/10.64898/2026.01.31.703034)
3. **Tissue Injury and Biomaterial Treatment Modulate Tumor Growth and Response to Immunotherapy.** Ruta, A., Gray-Gaillard, E.F., Garcia, J.A., Davenport Huyer, L., Mathkour, Y., Cherry, C., Patatanian, M., Mejias, J.C., Maestas Jr, D.R., Krishnan, K., Abraham, P., Wolf, M.T., Pardoll, D.M., and Elisseeff, J.H., *bioRxiv*, 2026. DOI: [10.64898/2026.02.02.703323](https://doi.org/10.64898/2026.02.02.703323)
4. **Regulatory T cells clonally expand and contribute to stromal cell function in fibrotic response to synthetic implants.** Mejías, J.C., Krishnan, K., Ruta, A., Ramanujam, A.S., Gray-Gaillard, E.F., Huyer, L.D., Maestas, D.R., Nagaraj, S., Yu, F.H., Rindone, A.N., Abraham, P., Browne, M., Fertig, E.J., Pardoll, D.M., and Elisseeff, J.H., *bioRxiv*, 2026. DOI: [10.64898/2026.01.05.697727](https://doi.org/10.64898/2026.01.05.697727)

PEER-REVIEWED PUBLICATIONS

1. **Helminth Egg Derivatives as Pro-Regenerative Immunotherapies.** Maestas Jr, D.R., Chung, L., Han, J., Wang, X., Sommerfeld, S.D., Kelly, S.H., Moore, E., Nguyen, H.H., Mejías, J.C., Peña, A.N. and Zhang, H., 2023. *Proceedings of the National Academy of Sciences*, 120(8), p.e2211703120. DOI: [10.1073/pnas.2211703120](https://doi.org/10.1073/pnas.2211703120)
2. **Long-term remodeling of compliance-matched tissue engineered vascular grafts is sex specific.** Martinet, K.M., Maestas, D.R. Jr., Kohyama, K., Min, L.X., Gholami, R., Avila, J.L., Ye, S.H., Kim, K., Wagner, W.R. and Vande Geest, J.P., *Acta Biomaterialia*, 2026. DOI: [10.1016/j.actbio.2026.06.042](https://doi.org/10.1016/j.actbio.2026.06.042)
3. **γδ T cell-stromal networks modulate matrix composition and vascularity in foreign body response.** Ruta, A., Krishnan, K., Woo, J., Mejías, J. C., Gray-Gaillard, E. F., Maestas Jr, D.R., & Elisseeff, J. H. *Nature Communications* (2026). DOI: [10.1038/s41467-026-71540-7](https://doi.org/10.1038/s41467-026-71540-7)
4. **Innate and Adaptive Immune Responses to Clinical Hyaluronic Acid Fillers.** Hooks, J.S., Yang, B., Maestas Jr, D.R., Andorko, J.I., Ruta, A., Mejias, J.C., Kelly, S.H., Hee, C.K. and Elisseeff, J.H., 2025. *Journal of cosmetic dermatology*, 24(7), p.e70292. DOI: doi.org/10.1111/jocd.70292
5. **Murine gut microbiota dysbiosis via enteric infection modulates the foreign body response to a distal biomaterial implant.** Yang, B., Rutkowski, N., Ruta, A., Gray-Gaillard, E., Maestas Jr, D.R., Kelly, S.H., Krishnan, K., Wu, X., Wu, S., Chen, A. and Mejías, J.C., 2025. *Proceedings of the National Academy of Sciences*, 122(20), p.e2422169122. DOI: doi.org/10.1073/pnas.2422169122
6. **Age-associated Senescent-T Cell Signaling Promotes Type 3 Immunity that Inhibits the Biomaterial Regenerative Response.** Han, J., Cherry, C., Mejías, J.C., Krishnan, K., Ruta, A., Maestas Jr, D.R., Peña, A.N., Nguyen, H.H., Nagaraj, S., Yang, B. and Gray-Gaillard, E.F., 2024. *Advanced Materials*, 36(43), p.2310476. DOI: [10.1002/adma.202310476](https://doi.org/10.1002/adma.202310476)
7. **Improving the hemocompatibility of a porohyperelastic layered vascular graft using luminal reversal microflows.** Behrangzade, A., Ye, S.H., Maestas Jr, D.R., Wagner, W.R. and Geest, J.P.V., 2024. *Journal of the mechanical behavior of biomedical materials*, 157, p.106638. DOI: [10.1016/j.jmbbm.2024.106638](https://doi.org/10.1016/j.jmbbm.2024.106638)
8. **Design of cell-type-specific hyperstable IL-4 mimetics via modular de novo scaffolds.** Yang, H., Ulge, U.Y., Quijano-Rubio, A., Bernstein, Z.J., Maestas, D.R., Chun, J.H., Wang, W., Lin, J.X., Jude, K.M., Singh, S. and Orcutt-Jahns, B.T., 2023. *Nature chemical biology*, pp.1-11. DOI: [10.1038/s41589-023-01313-6](https://doi.org/10.1038/s41589-023-01313-6)
9. **Creation and preclinical evaluation of a novel mussel-inspired, biomimetic, bioactive bone graft scaffold: direct comparison with Infuse bone graft using a rat model of spinal fusion.** Cottrill, E., Pennington, Z., Wolf, M.T., Dirckx, N., Ehresman, J., Perdomo-Pantoja, A., Rajkovic, C., Lin, J., Maestas, D.R., Mageau, A. and Lambrechts, D., *Journal of Neurosurgery: Spine*, 2023. DOI: [10.3171/2023.2.SPINE22936](https://doi.org/10.3171/2023.2.SPINE22936)
10. **Autologous Protein Solution processing alters lymphoid and myeloid cell populations and modulates gene expression dependent on cell type.** Pena, A.N., Sommerfeld, S.D., Anderson, A.E., Han, J., Maestas, D.R., Mejias, J.C., Woodell-May, J., King, W., Ganguly, S., Elisseeff, J.H., *Arthritis Research & Therapy*, 2022. DOI: [10.1186/s13075-022-02875-x](https://doi.org/10.1186/s13075-022-02875-x)
11. **Biomaterials-based immunomodulation enhances survival of murine vascularized composite allografts.** Sommerfeld, S.D., Zhou, X., Mejías, J.C., Oh, B.C., Maestas, D.R., Furtmüller, G.J., Laffont, P.A., Elisseeff, J.H. and Brandacher, G., *Biomaterials science*, 2023. DOI: [10.1039/D2BM01845D](https://doi.org/10.1039/D2BM01845D)
12. **Biomaterials direct functional B cell response in a material specific manner.** Moore, E., Maestas, D.R., Cherry, C.C., Garcia, J.A., Comeau, H.Y., Davenport Huyer, L., Kelly, S., Pena, A.N., Blosser, R., Rosson, G., Elisseeff, J.H., *Science Advances*, 2021. DOI: [10.1126/sciadv.abj5830](https://doi.org/10.1126/sciadv.abj5830)
13. **An immunologically active, adipose-derived extracellular matrix biomaterial for soft tissue reconstruction: concept to clinical trial.** Anderson, A.E., Parrillo, A.J., Wolf, M.T., Maestas, D.R., Graham, I.,

- Tam, A., Payne, R., Aston, J., Cooney, C., Byrne, P., Cooney, D., Elisseeff, J.H., 2021. *npj Regenerative Medicine*. DOI:[10.1038/s41536-021-00197-1](https://doi.org/10.1038/s41536-021-00197-1)
14. **Computational reconstruction of the signaling networks surrounding implanted biomaterials from single-cell transcriptomics.** Cherry, C.C., **Maestas, D.R.**, Han, J., Andorko, J.I., Cahan, P., Fertig, E., Garmire, L.X., Elisseeff, J.H., *Nature Biomedical Engineering*, 2021. DOI:[10.1038/s41551-021-00770-5](https://doi.org/10.1038/s41551-021-00770-5)
 15. **Type 2 immunity induced by bladder extracellular matrix enhances corneal wound healing.** Wang, X., Chung, L., Hooks, J., **Maestas, D.R.**, Lebid, A., Andorko, J.I., Huleihel, L., Chin, A.F., Wolf, M., Remlinger, N.T., Stepp, M.A., *Science Advances*, 2021, 7(16), p.eabe2635. DOI: [10.1126/sciadv.abe2635](https://doi.org/10.1126/sciadv.abe2635)
 16. **The immune system and its contribution to variability in regenerative medicine.** Moore, E.M., **Maestas, D.R.**, Comeau, H.Y. and Elisseeff, J.H., *Tissue Engineering Part B: Reviews*, 2021. DOI: [10.1089/ten.teb.2019.0335](https://doi.org/10.1089/ten.teb.2019.0335).
 17. **IL-17 and immunologically induced senescence regulate response to injury in osteoarthritis.** Faust, H.J., Zhang, H., Han, J., Wolf, M.T., Jeon, O.H., Sadtler, K., Peña, A.N., Chung, L., **Maestas, D.R.**, Tam, A.J. and Pardoll, D.M., Elisseeff, J.H., 2020. *Journal of Clinical Investigation*, 130(10), pp.5493-5507. DOI: [10.1172/JCI134091](https://doi.org/10.1172/JCI134091).
 18. **Interleukin 17 and senescent cells regulate the foreign body response to synthetic material implants in mice and humans.** Chung, L., **Maestas, D.R.**, Lebid, A., Mageau, A., Rosson, G.D., Wu, X., Wolf, M.T., Tam, A.J., Vanderzee, I., Wang, X., Andorko, J.I., & Elisseeff, J.H., *Science translational medicine*, 2020. DOI: [10.1126/scitranslmed.aax3799](https://doi.org/10.1126/scitranslmed.aax3799).
 19. **Interleukin-36γ-producing macrophages drive IL-17-mediated fibrosis.** Sommerfeld, S.D., Cherry, C., Schwab, R.M., Chung, L., **Maestas, D.R.**, Laffont, P., Stein, J.E., Tam, A., Ganguly, S., Housseau, F. Taube, J.M., Elisseeff, J.H., *Science immunology*, 2019. DOI:[10.1126/sciimmunol.aax4783](https://doi.org/10.1126/sciimmunol.aax4783).
 20. **Surface modification of electrospun scaffolds for endothelialization of tissue-engineered vascular grafts using human cord blood-derived endothelial cells.** Ardila, D.C., Liou, J.J., **Maestas, D.R.**, Slepian, M.J., Badowski, M., Wagner, W.R., Harris, D. and Vande Geest, J.P., 2019. *Journal of clinical medicine*, 8(2), p.185. DOI: [10.3390/jcm8020185](https://doi.org/10.3390/jcm8020185).
 21. **Key players in the immune response to biomaterial scaffolds for regenerative medicine.** Chung, L., **Maestas, D.R.**, Housseau, F. and Elisseeff, J.H., 2017. *Advanced drug delivery reviews*, 114, pp.184-192. DOI: [10.1016/j.addr.2017.07.006](https://doi.org/10.1016/j.addr.2017.07.006).
 22. **2-Photon Characterization of Optical Proteolytic Beacons for Imaging Changes in Matrix-Metalloprotease Activity in a Mouse Model of Aneurysm.** Haskett, D.G., **Maestas, D.R.**, Howerton, S.J., Smith, T., Ardila, D.C., Doetschman, T., Utzinger, U., McGrath, D., McIntyre, J.O. Vande Geest, J.P., 2016. *Microscopy and Microanalysis*, 22(2), pp.349-360. DOI: [10.1017/S1431927616000088](https://doi.org/10.1017/S1431927616000088).

PATENTS AND PATENT APPLICATIONS

1. **Regenerative helminth containing compositions and uses thereof.** Elisseeff, J., Pardoll, D.M. and **Maestas, D.R.**, Johns Hopkins University, 2025. U.S. Patent Application 18/894,513.

CONFERENCE PRESENTATIONS, POSTERS, & PRESENTED WORKSHOPS

- **“Resolving differential vascular graft remodeling using longitudinal multiphoton tracking in a 3D culture platform,”** **Maestas, D.R.**, T.R. Murphy, K.M. Martinet, T. Moyston, L.X. Min, S. Ye, WR Wagner, JP Vande Geest. *UPPDA Annual Research Symposium*. Poster Presentation. Pittsburgh, PA. July 2026.
- **“Vascular Engineering Workshop,”** **Maestas, D.R.** in lieu of Vande Geest, J.P. Presented and ran a hands-on workshop focused on the design of modern cutting-edge vascular grafts. *Vasculata 2025: Vascular Bioengineering – Next Gen Systems*. Pittsburgh, PA. July 2025.
- **“2-Photon Intravital Imaging of Small Diameter Vascular Grafts in an Aortic 3D Culture System,”** **Maestas, D.R.**, Murphy, T.R., Ye, S., Wagner, W.R., Vande Geest, J.P., NIH Diversity Scholar Workshop. Poster Presentation. October 2024.
- **“Helminth Egg Antigens as Pro-Regenerative Immunotherapies,”** **Maestas, D.R.**, Chung, L., Sommerfeld, S.D., Pardoll, D.M., Elisseeff, J.H., *The Tumor Immunology Research in Progress. Podium Presentation, Baltimore, MD, 2021 and again in 2022.*
- **“Helminth Egg Antigens as Pro-Regenerative Immunotherapies,”** **Maestas, D.R.**, Chung, L., Sommerfeld, S.D., Elisseeff, J.H., *BMES Conference Podium Presentation, 2021.*
- **“Skeletal Muscle Regeneration Enhanced by Natural Immunomodulatory Scaffolds,”** **Maestas, D.R.**, Chung, L., Sommerfeld, S.D., Pardoll, D.M., Elisseeff, J.H., *BMES Conference Podium Presentation, Atlanta, GA, 2019.*
- **“Tissue Engineered Vascular Grafts Seeded with Human Umbilical Cord Blood Derived Endothelial Cells,”** **Maestas, D.R.**, Ardilla, C., Harris, D.T., Slepian, M. Vande Geest, J.P. Paper and research poster presented at The Undergraduate Research Opportunities (URO) Consortium’s Summer Colloquium; abstract published in the UROC Abstract Review, The University of Arizona, Tucson, AZ, 2015.
- **“Automated Whole-Body Imaging Integrated with Dermoscopy Imaging,”** Poster presented at The University of Arizona Engineering Design Day Conference, University of Arizona, Tucson, AZ, 2015.

ACADEMIC LEADERSHIP ROLES

Site Head for UPMC Hillman Cancer Center Academy	2026-
Swanson School of Engineering Safety Committee Bioengineering Representative	2025-
Undergraduate and Graduate Student Mentor	2016-2021
BioTechnical Institute Internship Mentor (six times)	2016-2019
P-TECH Advisory Council and Student Mentor	2017-2020
P-TECH BME Student Recruiter, Team Lead	2016-2020
BME PhD Recruitment Big Sibling	2016-2021
JHU BME PhD Student Council Co-President	2017-2018
JHU BME PhD Student Recruitment Team Leader	2016-2017
Mathematics Instructor: Upward Bound (TRiO), Pima Community College	2013-2016
CRLA Certified Master Tutor, Trainer, Pima Community College	2012-2016

EMPLOYMENT EXPERIENCES

Mathematics Instructor – <i>Pima Community College</i>	2014–2015
Upward Bound, TRiO program. Math instructor and tutor for high school students	
College Learning Center Tutor – <i>Pima Community College</i>	2013–2015
Pima Community College East. CRLA Certified Level III Master Tutor for writing, Calculus III & Differential Equations, General & Organic Chemistry, & Physics	
General Manager – <i>Wendy's Old-Fashioned Hamburgers</i>	2008–2012
Full store management overseeing restaurant operations, food safety, cleanliness, customer service, daily operations. Phase II Training Store Certified.	

RESEARCH TECHNICAL EXPERTISE

In Vivo Injury and Surgical Models: Rat and murine handling; aseptic surgical technique; volumetric muscle loss, ACL transection, corneal laceration, biomaterial implantation, intraperitoneal/tail-vein/intramuscular/retro-orbital injections, gavage, cheek bleeding, wound care, controlled substance procedures, and functional testing including ACLT and skeletal muscle treadmill assays.

Biomaterials, Tissue Engineering, and Scaffold Fabrication: Decellularized ECM production from small intestine submucosa, urinary bladder, cardiac muscle, and skeletal muscle; decellularization verification; electrospinning; hydrogel and vitrified hydrogel fabrication; 3D scaffold production for cell culture; helminth soluble egg antigen isolation; 3D printing.

Microscopy, Histology, and Imaging: Scanning electron microscopy, two-photon microscopy, confocal microscopy, EVOS and Zeiss AxioImager imaging; cryostat and FFPE microtomy; FFPE embedding & processing; histological staining including Masson's trichrome, Picrosirius Red, Safranin-O, IHC, IF, ICC, and immunofluorescence imaging.

Molecular, Cellular, and Immunological Assays: Primary cell isolation and culture, stem cell differentiation, cryopreservation, PCR/qRT-PCR, SDS-PAGE and Western blotting, ELISA, NanoString, gel zymography, MTS/MTT assays, platelet activity assays, flow cytometry, FACS, and MACS-based cell sorting.

Biomechanics and Specialized Equipment: Biaxial loading bioreactor, ultracentrifugation, Qubit fluorimetry, cryomilling, lyophilization, thermocyclers, autoclave/glassware washer systems, cryofreezer and LN₂ dewar management.

Data Analysis and Software: MATLAB, GraphPad Prism, ImageJ/FIJI, FlowJo, nCounter/Rosalind, QuantStudio and StepOne analysis software, Zeiss ZEN, Microsoft Office, R, Python, and entry-level SolidWorks.

AWARDS AND RECOGNITIONS

- **NHLBI Supplement Scholar to Parent R01** — *University of Pittsburgh*

The NHLBI Diversity Supplement Award provides research support to permit individuals in the postdoctoral phase of their training to participate in ongoing research projects and career development experiences in preparation for an independent career in health-related research.

2023-2025

- **P-TECH Outstanding Mentorship Team** — *Johns Hopkins*

Award to recognize the largest cohort and contribution of mentors, tutors, and participants recruited to volunteer for the community P-TECH program.

2019

- **National Science Foundation Graduate Research Fellow** — *Johns Hopkins*

The National Science Foundation. Graduate research fellowship awarded to students who demonstrate potential to strengthen the vitality of the U.S.A.'s science and engineering enterprise.

2017-2022

- **Drescher Graduate Medical Research Award** — *Johns Hopkins*

Johns Hopkins University. Recognition award for outstanding first-year graduate students engaged in biomedical research.

2017

- **Galileo Circle Scholar Student Award** — *The University of Arizona*

A scholarship awarded to undergraduate students who demonstrate exceptional potential in the physical, mathematical, environmental, cognitive, or life sciences.

2016

- **BIO5 Institute Innovator Award** — *The BIO5 Institute*

The University of Arizona. Awarded at The Undergraduate and Professional Student Council Showcase. Recognizes a student exhibitor whose work most embodied multidisciplinary research, innovation in education & entrepreneurship.

2016

- **Danny Brower Memorial Scholar** — *The University of Arizona*

The University of Arizona. A scholarship awarded to students who demonstrated strong research experience and demonstrated financial need.

2015

- **Highest Academic Distinction** — *The University of Arizona*

Awarded to students with more than 30 units in an academic year and attained a 4.0 grade point average during this period.

2014-2015

- **Dean's List with Distinction** — *The University of Arizona*

Awarded to students who attained a 4.0 grade point average.

2014-2015

- **Tutor of the Year-Pima East** — *Pima Community College*

Named outstanding tutor of the year for the Pima College East Learning Center.

2013

- **Art Alberdine Math Scholar Award** — *Pima Community College*

A Pima Community College East Campus Scholarship award for students demonstrating high skill and potential with mathematics.

2012

- **Board of Governor's Special Recognition** — *Pima Community College*

A Pima Community College Special Recognition Award for students demonstrating outstanding performance at the two-year college level.

2012

- **AMATYC Award** — *Pima Community College*

American Mathematical Association of Two-Year Colleges Math Scholar Award. Award for outstanding performance on the annual student mathematics league.

2011 & 2012